

Software Requirements Specification (SRS)

System: Abajifar — CARAVAN Lounge Restaurant Management System
Document: SRS v1.1 · **Updated:** August 20, 2026
Standard: Structured after IEEE 830

1. Introduction

1.1 Purpose

This SRS specifies the complete requirements — functional and non-functional — for the Abajifar Restaurant Management System serving CARAVAN Lounge. It is intended for developers, testers, and stakeholders taking over or extending the system.

1.2 Scope

The system digitizes end-to-end restaurant operations: table-side ordering (POS), kitchen workflow, cashiering, inventory and procurement, staff and branch administration, and management analytics. It is a web application accessed from browsers on desktops and tablets.

1.3 Definitions

Term	Meaning
POS	Point of Sale — the screen waiters use to compose orders.
Order Board	Live kanban-style view of order progress.
JWT	JSON Web Token used for stateless authentication.

Item Request	An internal request for stock items from the store.
--------------	---

2. Overall Description

2.1 Product Perspective

Three-tier logical architecture: a **Next.js 15 / React 19 frontend**, a **NestJS REST API backend**, and a **PostgreSQL database** accessed through TypeORM. The registered Abajifar web artifact runs the actual source in `artifacts/nextjs-app`. The frontend calls the backend through `/backend/api/*`; Next.js rewrites requests to the backend's `/api/*`. The separate `artifacts/api-server` is not the restaurant API.

2.2 User Classes

Eight roles: admin, owner, manager, coordinator, waiter, chef, cashier, storekeeper (see FRS §2). Users are non-technical restaurant staff; the UI must remain simple, large-touch-friendly, and in plain language.

2.3 Operating Environment

Layer	Technology
Frontend	Next.js 15 (App Router), React 19, TypeScript, Tailwind CSS, Zustand (auth store), Recharts, Axios
Backend	NestJS 10, TypeScript, Passport-JWT, bcrypt, TypeORM
Database	PostgreSQL 14+ (TypeORM <code>synchronize: true</code> in development)
Runtime	Node.js 18+, pnpm monorepo workspaces

2.4 Constraints

- Protected API access requires JWT. Login and health are public; seed operations are administrative.
- The current backend still enables TypeORM schema synchronization. Production is blocked until synchronization is disabled there and a tested migration/rollback baseline exists.

- Realtime behavior is implemented by 10-second polling, not WebSockets.
- Currency is Ethiopian Birr (ETB); demo data uses Ethiopian dishes and names.

3. Functional Requirements

The complete functional catalogue is maintained in the FRS (document 01). Summary of subsystems:

Subsystem	Capabilities
Auth	Login, JWT issuance, role-based menus and guards
Orders / POS	Create, append items, confirm + assign chef, status lifecycle, filters, auto-refresh
Kitchen	Chef queues, accept/preparing/ready, live progress board with delay flags
Payments	Cash/card/wallet payment capture, change, daily report, shifts
Menu	Categories and items with availability
Inventory	Stock, low-stock alerts, suppliers, purchase orders, adjustments, item requests
Admin	Staff, restaurants, branches, tables
Analytics	Manager/owner one-page dashboard, exportable reports (Excel/PDF), cashier-scoped reports
Notifications	Role/user-targeted messages with read tracking

4. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	

		Standard pages should render within 2 s on a typical broadband connection; list endpoints return within 500 ms for up to 10,000 orders.
NFR-02	Freshness	Operational views (orders, kitchen, boards, dashboards) auto-refresh every 10 seconds without flicker.
NFR-03	Security	JWT-based auth; bcrypt password hashing; role guards on every protected endpoint; secrets provided via environment variables, never committed.
NFR-04	Usability	Plain-language UI, color-coded statuses, single-screen layouts that minimize scrolling; usable on tablets (≥ 768 px).
NFR-05	Reliability	Backend failures surface explicit errors; the frontend degrades gracefully (empty states, retry on next poll).
NFR-06	Maintainability	TypeScript end-to-end; modular NestJS feature modules; monorepo with shared conventions; <code>tsc --noEmit</code> must pass.
NFR-07	Portability	Runs anywhere Node 18+ and PostgreSQL are available; deployable to Replit, Vercel (frontend) + managed Node host (backend), or a single VM.
NFR-08	Auditability	Stock adjustments and item requests record acting users and timestamps.

5. External Interface Requirements

5.1 User Interface

Single-page-app feel with a persistent left sidebar (role-filtered), light gray theme with brand-orange accents (#E8832A), and modal dialogs for POS, payment, order detail, and confirmations.

5.2 API Interface

REST over HTTPS with JSON bodies. The browser-facing path is `/backend/api`; the backend's direct prefix is `/api`. Interactive Swagger documentation exists at backend path `/api/docs` and must be restricted or disabled in production. Main resource groups are `/auth`, `/users`, `/restaurants`, `/branches`, `/menus`, `/tables`, `/orders`, `/kitchen`, `/payments`, `/inventory`, `/notifications`, `/summary`, `/seed`.

5.3 Database Interface

PostgreSQL via TypeORM repositories; connection string in `DATABASE_URL`; SSL honored when `sslmode=require` is present.

6. Assumptions & Dependencies

- A single restaurant company operates the system; multiple branches are supported.
- Kitchen staff record stock deductions/waste for profitability estimates to be accurate.
- Exports (Excel/PDF) are generated client-side (xlsx, jsPDF) and require a modern browser.

7. Acceptance and Production Evidence

Architecture assessment is complete, but repository evidence for the full safe-refactoring, separation, automated-regression, security-hardening, migration, backup/restore, and production-verification stages is incomplete. Requirements marked in this SRS are acceptance targets, not automatic claims of production readiness. Refer to document 12 for current READY / NEEDS ATTENTION / BLOCKED status.